USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [22CSE330]

FOURTH SEMESTER B.Tech EXAMINATION, MAY-2024

THEORY OF COMPUTATION

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Write the purpose of extended transition function.	L2	CO1	**02**
	b.	List any four closure properties of regular language.	L1	CO2	**02**
	c.	Write the formal definition of Context Free Grammar	L1	CO3	**02**
	d.	What is the need for PDA?	L2	CO5	**02**
	e.	Draw the Turing Machine Model.	L2	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Differentiate DFA and NFA with formal definition	L2	CO2	**05**
		OR			
	b.	An IT company introduce new password rules to their employees, So that all the employees asked to set a new password for their official login which starts with the character "ba" followed by any number of a's and b's in any combination. Design a DFA which accepts the passwords which follows above rules.	L2	CO1	**05**
3.	**a.**	Prove that regular languages are closed under intersection.	L2	CO2	**05**
		OR			
	b.	State pumping Lemma.	L2	CO2	**05**
4.	**a.**	Construct CFG for the Language L={WcW^R / W belongs to $(a+b)^*$ and c is constant. Note: W^R means reverse of W.	L3	CO3	**05**
		OR			
	b.	In a CFG the production rules are S->aCa C->aCa **|** b Find the language generated by the CFG.	L3	CO3	**05**
5.	**a.**	Consider the CFG E->E+E | E*E | id Find the left most derivation and right most derivation for the string id+id*id	L3	CO4	**05**
		OR			
	b.	Consider the CFG E->E+E | E*E | id Draw two different parse tree for the string id+id*id	L3	CO4	**05**

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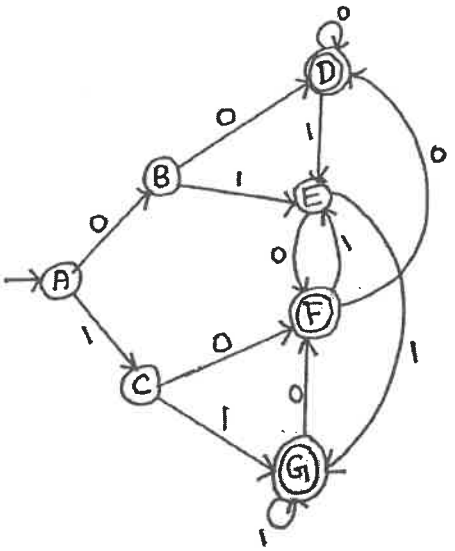
6.	**a.**	Explain the two ways to accept a language by a PDA.	L3	CO5	**05**
		OR			
	b.	Design a PDA which accepts the language $L=(a^n b^n / n>=1\}$.	L3	CO5	**05**
7.	**a.**	Design Turing machine to compute x+1.	L3	CO5	**05**
		OR			
	b.	Design Turing machine to compute x+2.	L3	CO5	**05**
		PART-C			**6 X 10=60**
8.	**a.**	Consider the language L= {W/W has even number of 1's alone or even number of 0's alone or both even number of 0's and even number of 1's } Design a DFA which accepts the strings that satisfies any of the above condition and check whether the string "10110" is accepted by the DFA or not.	L2	CO1	**10**
		OR			
	b.	Construct the NFA accepting the language L={x/x ∈ (a,b)* where |x|=3 and the third symbol of x from the right end is b and check whether the string "aabab" is accepted by the NFA or not.	L2	CO1	**10**
9.	**a.**	Show that the following languages are not regular using pumping lemma. **i.** $L=\{a^p/P$ is a prime number$\}$ is not a regular. **ii.** $L=\{a^i b^j c^k / 0<=i<=j<k\}$ is not regular.	L2	CO3	**10**
		OR			
	b.	Minimize the following DFA using table filling algorithm. 	L2	CO2	**10**

10.	**a.**	Consider the below CFG productions S->I S->S+S S->S*S S->(S) I->a | b | Ia | Ib I-I1 | I0 (Note: I1, I0 are Numeric 1 and 0, |means or) For the string a*(a+b00) find Left most derivation and Right most derivation.	L3	CO3	**10**
		OR			
	b.	Consider the below CFG productions S->A1B A->0A | ∈ B->0B | 1B | ∈ Find the derivation and Parse Tree for the yields a. 00101 b. 00011	L3	CO3	**10**
11.	**a.**	Convert the following CFG into PDA and test whether the string E-> E+E | E*E | a and test whether the string a*a+a is accepted by the PDA or not.	L3	CO5	**10**
		OR			
	b.	Convert the following PDA into CFG $\delta(q_0, 0, z_0) = \{(q_0, x z_0)\}$ $\delta(q_1, 1, x) = \{(q_1, \varepsilon)\}$ $\delta(q_0, 0, x) = \{(q_0, xx)\}$ $\delta(q_1, \varepsilon, x) = \{(q_1, \varepsilon)\}$ $\delta(q_0, 1, x) = \{(q_1, \varepsilon)\}$ $\delta(q_1, \varepsilon, z_0) = \{(q_1, \varepsilon)\}$	L3	CO5	**10**
12.	**a.**	Design a PDA which accepts the language L={WW^R / W in (0+1)*}	L3	CO5	**10**
		OR			
	b.	Design a PDA which accepts the language L={$a^n b^m a^n$ / m,n>=1} by accepting empty stack	L3	CO5	**10**
13.	**a.**	Design Turing machine for the language L={$0^n 1^n$ / n>=1}	L3	CO5	**10**
		OR			
	b.	Explain the various programming techniques for Turing machine Construction.	L3	CO5	**10**